Staphylococcus aureus Pneumonia — Comprehensive Disease Characteristics Report

Disease: Staphylococcus aureus Pneumonia **MONDO ID:** MONDO:0041879 **Category:** Infectious Disease **Report basis:** 5-iteration autonomous literature investigation; 16 confirmed findings; 69 papers reviewed.

Summary

Staphylococcus aureus pneumonia is a severe, frequently life-threatening bacterial lower-respiratory-tract infection caused by the Gram-positive coccus *Staphylococcus aureus* (NCBI:txid1280), spanning two dominant clinical contexts: (1) **nosocomial pneumonia** — hospital-acquired (HAP) and ventilator-associated pneumonia (VAP), where *S. aureus* is one of the top pathogens, and (2) **community-acquired necrotizing pneumonia**, an aggressive syndrome in young, previously healthy hosts driven by the Panton-Valentine leukocidin (PVL). Unlike the genetic disorders this template is often applied to, *S. aureus* pneumonia is an **infectious, non-Mendelian disease**: there are no causal human germline genes. Instead, the "genetics" of the disease reside in the **bacterial virulence gene repertoire** (*hla*, *lukS/F-PV*, *psmα*, *agr*), while host genetic contributions are limited to immunodeficiency states (e.g., STAT3/Th17 defects) that raise susceptibility.

The mechanistic core of the disease is now well defined. The pore-forming cytotoxin **alpha-hemolysin (Hla)** binds its host receptor, the zinc-dependent metalloprotease **ADAM10**, on alveolar epithelial cells, upregulating ADAM10 protease activity that cleaves the adherens-junction protein **E-cadherin**, disrupting the epithelial barrier and producing lethal acute lung injury. In parallel, the **agr quorum-sensing** system drives expression of **phenol-soluble modulins (PSMα)** and PVL, which lyse neutrophils (PSMα via MLKL-dependent necroptosis), amplifying inflammation and tissue destruction. Antecedent **influenza** infection is a critical predisposing factor: type I interferon suppresses protective **Th17/IL-17/IL-22 and IL-1β** immunity, converting mild colonization into fulminant necrotizing disease.

Clinically, the disease ranges enormously in severity. PVL-positive necrotizing community-acquired pneumonia carries 40–75% mortality with rapid deterioration, hemoptysis, leukopenia, and septic shock, whereas VAP has an attributable mortality of 3–17%. **PVL-positivity (HR 1.81) and methicillin resistance (HR 2.37) are independent predictors of mortality.** First-line therapy is **vancomycin or linezolid** for MRSA and **cefazolin or anti-staphylococcal penicillins** for MSSA, with toxin-suppressing antibiotics (linezolid, clindamycin) and anti-Hla monoclonal antibody prophylaxis (**suvratoxumab/MEDI4893**) as emerging strategies. No licensed *S. aureus* vaccine exists.

Section 1 — Disease Information

Overview. *S. aureus* pneumonia is inflammation and consolidation of the lung parenchyma caused by infection with *S. aureus*. It presents as: community-acquired pneumonia (CAP, often necrotizing and PVL-associated), hospital-acquired pneumonia (HAP), ventilator-associated pneumonia (VAP), and post-influenza secondary bacterial pneumonia. Nosocomial *S. aureus* pneumonia is a leading cause of ICU morbidity and mortality; the 2026 review of HAP/VAP found "*Pseudomonas aeruginosa* and *Staphylococcus aureus* predominate" (PMID: 42203422).

Key identifiers. - **MONDO:** MONDO:0041879 - **MeSH:** Pneumonia, Staphylococcal (D011026) - **ICD-10:** J15.211 (Pneumonia due to MRSA), J15.212 (Pneumonia due to MSSA); J15.2 (Pneumonia due to staphylococcus) - **ICD-11:** infectious pneumonia codes with *S. aureus* as causative organism - **SNOMED CT:** 53084003 (Bacterial pneumonia); staphylococcal pneumonia concept - **OMIM/Orphanet:** Not applicable — this is an acquired infectious disease, not a Mendelian or rare genetic disorder.

Synonyms / alternative names. Staphylococcal pneumonia; MRSA pneumonia; MSSA pneumonia; *S. aureus* necrotizing pneumonia; PVL-associated necrotizing pneumonia; CA-MRSA pneumonia; staphylococcal lung infection.

Information source type. The evidence base is **aggregated disease-level** (surveillance cohorts, ICU registries, case-control studies, systematic reviews, autopsy series, and animal-model mechanistic work), supplemented by **individual patient** case reports for rare fulminant presentations.

Section 2 — Etiology

Primary cause (infectious). The disease is caused by *Staphylococcus aureus*, including both methicillin-susceptible (MSSA) and methicillin-resistant (MRSA) strains, and both community-associated (CA-MRSA, e.g., USA300) and hospital-associated (HA-MRSA) lineages. Virulence is determined by the bacterial toxin repertoire rather than host genetics.

Risk factors. - **Host/environmental:** mechanical ventilation and endotracheal intubation (VAP), ICU admission, prior hospitalization, aspiration (AOR 4.43, 95% CI 1.74–11.24 for pneumonia-pathogen presence, PMID: 41557657), older age, immunosuppression, cancer, liver dysfunction, heart failure, long-term catheterization, and high SOFA/qSOFA scores (PMID: 41758741). **Antecedent influenza (or influenza-like illness)** is a major predisposing factor for necrotizing CA disease. Nasal/nasopharyngeal *S. aureus* colonization is a reservoir. Occupational exposure to livestock (LA-MRSA CC398) increases carriage. - **Genetic host susceptibility:** rare. Patients with **hyper-IgE (Job) syndrome** carrying **STAT3** mutations, "who often present with *S. aureus* infections of the lung and skin," lack Th17 immunity (PMID: 21178015). IL-1R1 and IL-17RA deficiencies raise susceptibility in models. - **Pathogen genetic risk factors:** carriage of *lukS-PV/lukF-PV* (PVL), high-level *hla* expression, functional *agr* locus, and *psma* operon.

Protective factors. - **Host lipoproteins:** serum LDL/ApoB sequesters the agr autoinducing peptide (AIP). "Activation of agr... is inhibited by apolipoprotein B, the structural protein of low-density lipoprotein, which binds and sequesters the secreted agr-signaling peptide (AIP)" (PMID: 26608923); lipoprotein deficiency worsened agr-dependent pneumonia. - **Intact Type 17 immunity** (IL-17/IL-22/IL-23) and IL-1β signaling. - **Influenza vaccination** (reduces the antecedent viral trigger).

Gene–environment interactions. The dominant interaction is **virus–bacterium**: influenza-induced type I IFN suppresses host Th17/IL-1β responses (PMID: 21178015, PMID: 24089191), permitting toxin-armed *S. aureus* to cause necrotizing disease — a bacterial-genotype × viral-environment interaction.

Section 3 — Phenotypes

Phenotype	Type	HPO suggestion	Frequency / severity	Notes
Fever (>39 °C)	Symptom/ sign	HP:0001945 (Fever)	Common; higher temp in PVL+ (p=0.01)	PMID: 11888586
Cough / dyspnea / respiratory distress	Symptom	HP:0012735, HP:0002094	Very common	Respiratory distress 75% PVL+
Hemoptysis / pulmonary hemorrhage	Sign	HP:0002105 (Hemoptysis)	PVL+ 38% vs 3% (p=0.005)	Hallmark of necrotizing disease
Tachycardia (>140 bpm)	Sign	HP:0001649 (Tachycardia)	PVL+ enriched (p=0.02)	
Leukopenia	Lab abnormality	HP:0001882 (Leukopenia)	PVL+ ~79%; predicts death (p=0.001–0.002)	Poor-prognosis marker
Septic shock / hypotension	Sign	HP:0031273 (Septic shock)	Frequent in fatal PVL+ cases	PMID: 42501530
Pleural effusion / empyema	Physical/ imaging	HP:0002202, HP:0100751	PVL+ effusion onset (p=0.004)	Complication
Hypoxemic respiratory failure / ARDS	Sign	HP:0002878	Common in severe disease	PMID: 42454265

Age of onset. Bimodal. PVL-positive necrotizing CAP strikes **young, previously healthy** patients (median age 14.8 yr vs 70.1 yr for PVL-negative, p=0.001; PMID: 11888586); nosocomial disease predominates in older, comorbid, ventilated adults.

Severity/progression. Acute, rapidly progressive. In fatal PVL cases, time from onset to death was 5.5 ± 10.1 days (PMID: 21051855); 48-hour survival was 63% (PVL+) vs 94% (PVL−) (p=0.007). Episodic/relapsing patterns are not typical; the disease is self-limited with cure or progresses to death.

Quality of life. Survivors of severe ICU pneumonia experience post-intensive-care morbidity (muscle wasting, cognitive impairment) common to critical illness; disease-specific QoL instruments (EQ-5D/SF-36) are not established for this entity.

Section 4 — Genetic/Molecular Information

This is an infectious disease — there are no causal HUMAN germline genes. The molecular determinants are **bacterial virulence genes**:

Bacterial gene	Product	Function	Evidence
hla	Alpha-hemolysin (Hla/α-toxin)	Pore-forming cytotoxin; binds ADAM10	PMID: 21926978, PMID: 38990361
lukS-PV / lukF-PV	Panton-Valentine leukocidin (PVL)	Bicomponent leukocidin; lyses neutrophils	PMID: 11888586
psmα1-4	Phenol-soluble modulins α	Leukocyte-lysis toxins; neutrophil necroptosis	PMID: 29500427
agr (RNAIII)	Accessory gene regulator	Quorum-sensing master virulence regulator	PMID: 29500427
seb, sec, tst	Superantigens SEB, SEC, TSST-1	T-cell superantigens driving toxic shock	PMID: 42161902
mecA (SCC*mec*)	PBP2a	Methicillin resistance	PMID: 23524477

Host genetic modifiers of susceptibility. *STAT3* (hyper-IgE syndrome; loss of Th17), *IL17RA*, *IL1R1* — loss-of-function increases susceptibility (human + mouse). HGNC: STAT3, IL17RA, IL1R1, ADAM10. **ADAM10** is the host receptor whose activity is subverted (not mutated).

Variant classification / allele frequency / somatic-germline. Not applicable in the ACMG/ClinVar sense — the disease is not inherited. "Variants" of clinical importance are **bacterial genotypes** (PVL+/−, MRSA/MSSA, clonal complexes CC398, CC5, CC8/USA300, CC30, ST80).

Epigenetics / chromosomal abnormalities. Not applicable to the human host. Bacterial global regulation is via *agr* quorum-sensing and *sae/sar* regulons.

Section 5 — Environmental Information

- **Environmental factors:** hospital/ICU environment, endotracheal tubes (biofilm reservoir — MRSA biofilm forms on ETTs, PMID: 30093624), contaminated surfaces, livestock/meat exposure.
- **Lifestyle factors:** injection drug use (associated with MSSA bacteremia/endocarditis), smoking (airway epithelial injury), crowding, and occupational livestock contact.
- **Infectious agents (the cause itself):** *Staphylococcus aureus* (NCBI:txid1280). **Co-infecting agents that potentiate disease:** Influenza A virus (H1N1) and Influenza B virus — the key viral co-pathogens converting *S. aureus* colonization into necrotizing/secondary pneumonia (PMID: 21178015, PMID: 41619830).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. *S. aureus* colonizes the nasopharynx/airway and is aspirated or inhaled into the lower respiratory tract → **leads to** alveolar bacterial burden. (Antecedent influenza injures respiratory epithelium and suppresses immunity — *inferred prerequisite* in young necrotizing cases.)
2. Rising bacterial density activates the **agr quorum-sensing** system (RNAIII) → **results in** upregulated secretion of Hla, PSMα, and PVL. (Demonstrated; agr blockade is protective.)
3. Secreted **Hla binds ADAM10** on alveolar epithelial cells → **upregulates ADAM10 metalloprotease activity → cleaves E-cadherin → disrupts adherens junctions and the epithelial barrier** → alveolar–capillary leak and lethal acute lung injury. (Demonstrated genetically: Adam10-conditional-knockout mice are resistant, PMID: 21926978.)
4. **Branch A — Neutrophil destruction:** PSMα1–3 induce **neutrophil necroptosis via MLKL phosphorylation** and PVL forms pores in leukocytes → **results in** leukopenia, release of DAMPs/LDH, and impaired bacterial clearance. (Demonstrated, PMID: 29500427.)
5. **Branch B — Immune subversion by prior virus:** Influenza type I IFN **suppresses IL-17/IL-22/IL-23 and attenuates IL-1β/NF-κB → impairs Type 17 antibacterial immunity →**

higher bacterial burden and mortality. (Demonstrated in IL-17R$^{-/-}$, IL-22$^{-/-}$, IL-1R1$^{-/-}$ mice, PMID: 21178015, PMID: 24089191.)

6. **Branch C — Inflammasome/pyroptosis amplification:** In influenza–MRSA co-infection, caspase-1 and **GSDMD-mediated macrophage pyroptosis** are activated, and dysregulated type I IFN enhances myeloid TNF-α while suppressing T-cell IFN-γ → **amplifies lung injury**. (Demonstrated; GSDMD inhibitor disulfiram ameliorated disease, PMID: 41927528, PMID: 42089309.)
7. Combined barrier failure + neutrophil lysis + cytokine storm → **necrotizing hemorrhagic bronchopneumonia**, hemoptysis, ARDS, septic shock → **culminates in** multi-organ failure and death in severe cases.

Pathway / process / cell-type detail

- **Molecular pathways:** agr quorum-sensing (RNAIII); ADAM10 sheddase pathway; NF-κB/IL-1β signaling; type I IFN (IFNAR1) signaling; IL-23→IL-17/IL-22 (Type 17) axis; MLKL necroptosis; caspase-1/GSDMD pyroptosis (inflammasome).
- **Cellular processes (GO):** GO:0006909 phagocytosis; GO:0070266 necroptotic process; GO:0070269 pyroptosis; GO:0006954 inflammatory response; GO:0098742 cell-cell adhesion via plasma-membrane adhesion molecules (E-cadherin cleavage); GO:0051817 modulation of host process by symbiont.
- **Protein dysfunction:** host **E-cadherin** proteolytic loss of function; **ADAM10** gain of pathological activity (usurped by Hla); pore formation in host membranes by Hla/PVL/PSM.
- **Immune involvement:** neutrophil-centric defense that is turned against the host (neutrophil necroptosis); Th17 deficiency; macrophage pyroptosis; superantigen-driven T-cell activation (SEB/SEC/TSST-1).
- **Tissue damage mechanisms:** membrane pore-formation/osmotic lysis, hemorrhagic necrosis, oxidative/inflammatory injury, barrier breakdown, edema.
- **Cell types (CL):** CL:0000775 neutrophil; CL:0000583 alveolar macrophage; CL:0002063 type II pneumocyte / CL:0002062 type I pneumocyte; CL:0000066 epithelial cell; CL:0000738 leukocyte.

Section 7 — Anatomical Structures Affected

- **Primary organ:** Lung (UBERON:0002048); lung parenchyma / alveoli (UBERON:0002299 alveolus).
- **Body system:** Respiratory system (UBERON:0001004).
- **Secondary/complication organs:** Pleura (UBERON:0000977) — parapneumonic effusion/ empyema; bloodstream — bacteremia; distant sites via metastatic infection; kidney/heart in multi-organ failure and endocarditis.
- **Tissues:** Alveolar epithelium (type I & II pneumocytes), bronchial epithelium, pulmonary capillary endothelium, connective tissue of interstitium.
- **Cells targeted:** alveolar epithelial cells (Hla/ADAM10/E-cadherin), neutrophils (PVL/PSMα), macrophages (GSDMD pyroptosis).
- **Subcellular compartments (GO CC):** plasma membrane (GO:0005886, pore formation); adherens junction (GO:0005912); extracellular space (secreted toxins); cytosol (inflammasome assembly).
- **Localization/lateralization:** Frequently **bilateral** in necrotizing disease ("bilateral necrotizing hemorrhagic bronchopneumonia," PMID: 42501530); multilobar involvement common in CA-MRSA (PMID: 21722847).

Section 8 — Temporal Development

- **Onset:** Acute to hyperacute. PVL-associated necrotizing CAP often follows an influenza-like prodrome by 1–several days, then deteriorates rapidly.
- **Progression:** Rapid. Fatal PVL cases: onset-to-death 5.5 ± 10.1 days; 48-h survival markedly reduced in PVL+ (63% vs 94%). VAP: median time to ventilator-associated condition onset ~6 days (PMID: 24709718).
- **Disease course pattern:** Acute, progressive, self-limited (resolution with effective therapy) or fatal; not relapsing-remitting.
- **Critical periods / intervention windows:** Early source control, prompt appropriate antibiotics, and toxin suppression within the first 24–72 h are decisive; anti-Hla prophylaxis targets the ventilated pre-symptomatic window.

Section 9 — Inheritance and Population

Inheritance: Not applicable — infectious, non-heritable. No Mendelian pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier-frequency parameters apply to the host.

Epidemiology. - *S. aureus* is among the **top pathogens of nosocomial pneumonia** (PMID: 42203422); in Chinese HAP cohorts it was the 4th-ranked pathogen at 5.88% (PMID: 41758741). - **VAP incidence:** 2–16 episodes/1000 ventilator-days; attributable mortality 3–17%; *S. aureus* is a leading VAP pathogen with 50–80% MRSA in some series (PMID: 23524477). - In a CDC ventilator-associated-event cohort, *S. aureus* was **the most frequently isolated organism (29%)** (PMID: 24709718).

Population demographics. - **Age/sex:** PVL-necrotizing disease — young, previously healthy, often male; nosocomial disease — older comorbid ICU patients. - **Geography/variants:** PVL prevalence among MRSA varies 0–62% across regions (GCC scoping review, PMID: 42799093); CA-MRSA clones vary by region (USA300 in North America; ST80-IV, ST30, ST22 elsewhere). LA-MRSA CC398 clusters in high-livestock-density areas (18.6–23% of human MRSA in one German region, PMID: 23418434).

Section 10 — Diagnostics

Clinical/lab tests. - **Lower-respiratory culture** (definitive): endotracheal aspirate vs bronchoalveolar lavage (BAL). BAL-directed diagnostic protocols lowered VAP-attributable mortality to ~1% (PMID: 22310133). - **Blood cultures** for bacteremia; **pleural fluid** analysis for empyema. - **Molecular:** metagenomic next-generation sequencing (mNGS) identifies co-infections (e.g., PVL+ MSSA + influenza B) when conventional tests fail (PMID: 41619830); PCR for *mecA*, *lukS/F-PV*. - **CBC:** leukopenia is an ominous prognostic sign.

Stewardship diagnostic — MRSA nasal PCR. Nasal-swab PCR has excellent negative predictive value: "The MRSA PCR assay demonstrated 88.0% sensitivity and 90.1% specificity, with a positive predictive value of 35.4% and a negative predictive value of 99.2%" (PMID: 24277023). A negative swab safely de-escalates empiric vancomycin (PMID: 38028911); use caution after intranasal mupirocin decolonization (PMID: 32959059).

Imaging. Chest X-ray/CT: multilobar consolidation, cavitation/necrosis, effusion, pneumatoceles, abscess.

Genetic/omics diagnostics for the host: Not applicable. (Host WGS/WES/panels/karyotype/CMA have no role.) Pathogen genomics (WGS, spa-typing, MLST) is used for surveillance/typing.

Clinical criteria & differential. IDSA/ATS HAP/VAP and CAP guidelines. Differential: pneumococcal, *Pseudomonas*, *Klebsiella*, *Acinetobacter*, and viral pneumonias; distinguishing features of *S. aureus* necrotizing disease include rapid cavitation, hemoptysis, leukopenia, and antecedent influenza.

Section 11 — Outcome / Prognosis

Setting	Mortality	Source
PVL+ necrotizing CAP (older patients)	~47%	PMID: 33833037
PVL+ necrotizing CAP (toddlers)	~15%	PMID: 33833037
Necrotizing staph CAP (92-case review)	42.9%	PMID: 21051855
VAP (attributable)	3–17%	PMID: 23524477

Independent mortality predictors: "Mortality in older patients was predicted by PVL-positivity (hazard ratio (HR) 1.81, 95% CI 1.03–3.17) and methicillin resistance (HR 2.37, 95% CI 1.29–4.34)" (PMID: 33833037). Additional poor-prognosis markers: leukopenia (p=0.002), influenza (p=0.011), hemoptysis (p=0.024) (PMID: 21051855).

Complications: parapneumonic effusion, empyema, necrotizing pneumonia, lung abscess (local); bacteremia, metastatic infection, multi-organ failure, ARDS, DIC, death (systemic) (PMID: 32919518). Toxin-suppressing antibiotics improve outcome (p=0.007, PMID: 21051855).

Recovery: Survivors may recover fully but face post-ICU morbidity. Disease is self-limited with cure or rapidly fatal without effective therapy.

Section 12 — Treatment

Pharmacotherapy (first-line). - **MRSA pneumonia:** vancomycin OR linezolid — "Both linezolid and vancomycin are approved by USFDA and IDSA guidelines for the management of nosocomial pneumonia due to methicillin-resistant Staphylococcus aureus (MRSA)" (PMID: 36852899). Linezolid shows higher clinical cure (53–59%) and less nephrotoxicity than vancomycin (43–47%) in HAP/VAP (PMID: 38238670, PMID: 36852899); vancomycin penetrates alveolar fluid poorly (PMID: 21722847). NCIT: Vancomycin (C935), Linezolid (C1697). - **MSSA pneumonia/bacteremia:** cefazolin or anti-staphylococcal penicillins (nafcillin/oxacillin) — "cefazolin or antistaphylococcal penicillins should be the first-line treatment for methicillin-susceptible Staphylococcus aureus (MSSA) bacteraemia," with cefazolin causing less nephrotoxicity (PMID: 40349971). NCIT: Cefazolin (C341), Nafcillin, Oxacillin. - **Toxin suppression:** linezolid and clindamycin inhibit toxin synthesis; toxin-inhibiting regimens improved outcomes in necrotizing CAP (PMID: 21051855, PMID: 21722847). - **Alternatives/ resistant strains:** daptomycin (bloodstream, not lung — inactivated by surfactant), telavancin; MDRO guidance recommends linezolid or vancomycin for MRSA pneumonia (PMID: 42661422).

Advanced/experimental therapeutics. - **Anti-alpha-toxin monoclonal antibody — suvratoxumab (MEDI4893):** SAATELLITE phase 2 RCT evaluated its efficacy "in reducing the incidence of S aureus pneumonia in patients in the intensive care unit (ICU) who are on mechanical ventilation" (PMID: 33894131). - **Anti-Hla scFvs / multivalent nanobodies:** increased survival comparable to vancomycin in murine MRSA pneumonia (PMID: 42133074, PMID: 42161902); AAV-delivered anti-α-toxin mAbs protect in models (PMID: 41941484). - **Anti-virulence small molecules:** agrA inhibitor norwogonin (PMID: 38990361); ClpP inhibitor sesamin (PMID: 39805732); Hla inhibitors (Jingfang mixture, PMID: 39083107). - **Host-directed:** GSDMD inhibitor disulfiram ameliorated co-infection pneumonia (PMID: 41927528); innate immune priming with DDM (PMID: 41619354).

Surgical/interventional & supportive. Pleural drainage/chest tube for empyema; source control; prone positioning and lung-protective ventilation for ARDS; ECMO in refractory cases (though selected fulminant MSSA managed without ECMO, PMID: 42454265).

Pharmacogenomics. Not a major factor; vancomycin dosing is guided by renal function/ therapeutic drug monitoring rather than host genotype.

Section 13 — Prevention

- **Primary prevention:** VAP bundles (head-of-bed elevation, oral care, sedation minimization, subglottic suctioning), hand hygiene, aspiration precautions, and antimicrobial stewardship; **influenza vaccination** to remove the key viral trigger.
- **Secondary prevention:** MRSA nasal PCR screening for stewardship/de-escalation; active surveillance; MRSA decolonization (intranasal mupirocin) in high-risk/ICU settings.
- **Immunization: No licensed *S. aureus* vaccine exists** — "Despite decades of effort, vaccine development has been unsuccessful, highlighting the need for alternative approaches" (PMID: 41941484). Toxoid/nanotoxoid and multi-antigen vaccines protect mice (PMID: 37097076, PMID: 32197534) but none are approved.
- **Prophylaxis:** anti-Hla mAb (suvratoxumab) for ventilated, colonized ICU patients (PMID: 33894131).
- **Public health / One Health:** livestock-associated MRSA CC398 surveillance and occupational infection control (PMID: 23418434).

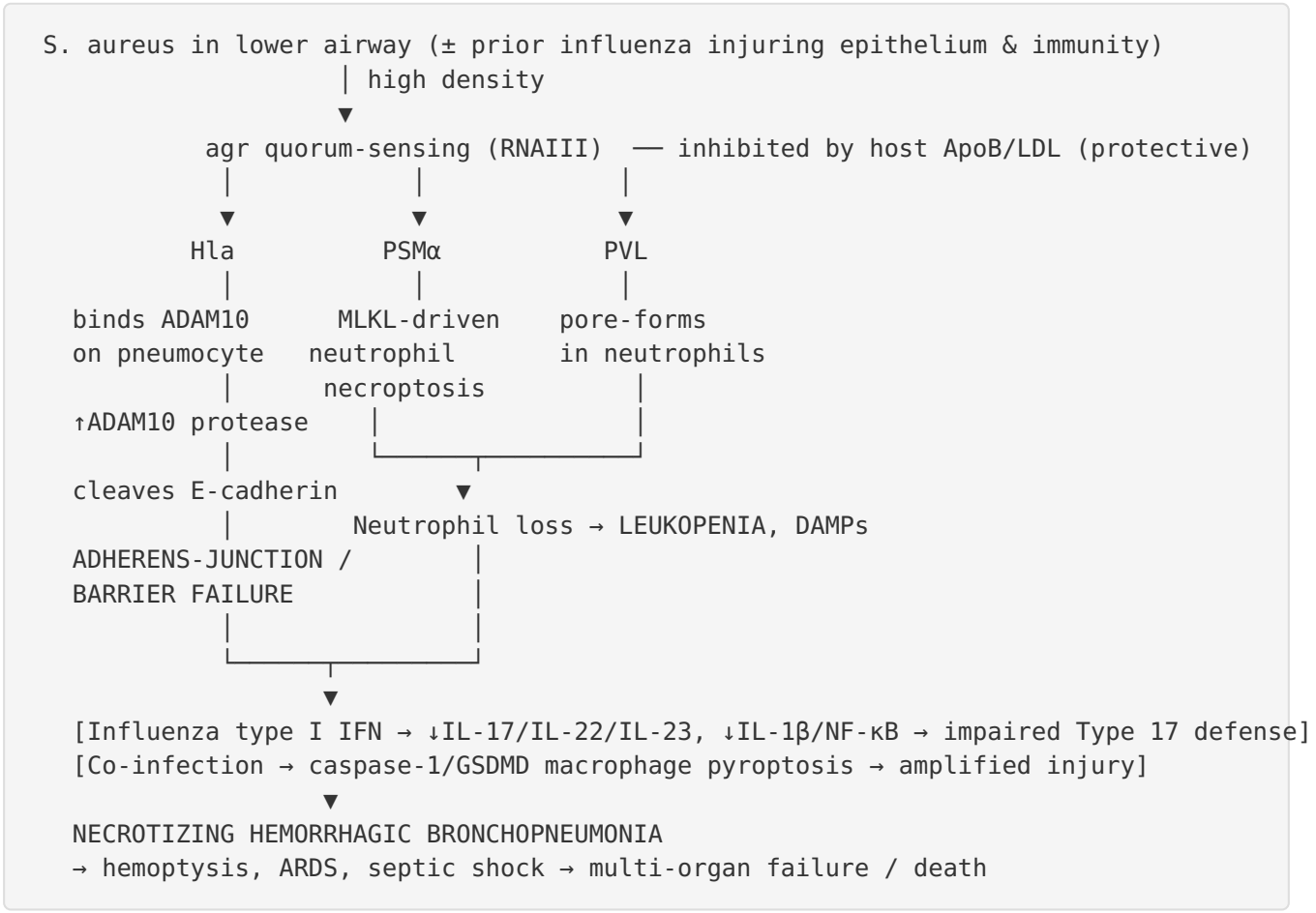
Section 14 — Other Species / Natural Disease

- **Taxonomy:** *Staphylococcus aureus* — NCBI:txid1280. Host species: human (NCBI:txid9606); reservoir/affected animals: pig (*Sus scrofa*, txid9823), cattle (*Bos taurus*, txid9913), poultry (chicken, txid9031).
- **Natural disease / veterinary relevance:** *S. aureus* causes bovine mastitis and infections across livestock; "Pigs, cattle and poultry are colonized with MRSA and the zoonotic transmission of such MRSA to humans via direct animal contact, environmental contaminations or meat are a matter of concern. Livestock-associated (LA) MRSA are mostly belonging to clonal complex (CC) 398" (PMID: 23418434). Other livestock lineages: CC5, CC9, CC97.
- **Zoonotic transmission:** direct animal contact, environmental contamination, and meat.
- **Comparative biology:** Hla/PVL toxin mechanisms are studied primarily in murine lungs; ADAM10 is conserved across mammals, supporting cross-species mechanistic relevance.

Section 15 — Model Organisms

- **Standard model:** murine **intranasal/intratracheal** *S. aureus* pneumonia, especially CA-MRSA **USA300** — the workhorse for pathogenesis and therapeutics (PMID: 42161902). Disease is "driven by potent virulence factors such as T-cell superantigens and alpha hemolysin," including SEB, SEC, and TSST-1.
- **Genetic models: Adam10 lung-epithelium conditional knockout** (resistant to lethal Hla pneumonia — landmark genetic proof, PMID: 21926978); **IL-17R$^{-/-}$, IL-22$^{-/-}$, IL-1R1$^{-/-}$** mice (impaired clearance, PMID: 21178015, PMID: 24089191); myeloid IFNAR1-deletion (improved survival, PMID: 42089309).
- **Co-infection models:** influenza A (PR8/H1N1) + MRSA sequential-infection mice (PMID: 41927528).
- **Large-animal: porcine** MRSA pneumonia models (ETT biofilm, telavancin/linezolid efficacy) (PMID: 30093624, PMID: 33046500).
- **In vitro:** A549 alveolar epithelial cells, RAW264.7 macrophages, primary human neutrophils.
- **Phenotype recapitulation:** murine models reproduce necrotizing pneumonia, toxin-mediated barrier injury, and neutrophil lysis well; **limitations** include species differences in PVL susceptibility (human neutrophils are far more PVL-sensitive than murine), immune-response divergence, and difficulty modeling the human influenza-priming context.

Mechanistic Model / Interpretation



Upstream vs downstream: agr activation and Hla–ADAM10 engagement are the **upstream** initiating molecular events; E-cadherin cleavage, neutrophil necroptosis, and immune subversion are **midstream**; hemorrhagic necrosis, ARDS, and shock are **downstream** clinical endpoints. The convergence of *epithelial barrier failure* (Hla/ADAM10) and *innate-effector destruction* (PVL/PSMα) plus *impaired adaptive/innate immunity* (post-influenza) explains why *S. aureus* pneumonia can be so rapidly lethal.

Evidence Base (key papers)

PMID	Contribution
21926978	**Landmark**: Hla→ADAM10→E-cadherin barrier mechanism; Adam10-cKO mice resistant
29500427	agr→PSMα→MLKL neutrophil necroptosis; agr blockade protective
26608923	Host ApoB/LDL sequesters agr AIP — protective factor
21178015	Influenza type I IFN suppresses Th17; STAT3/hyper-IgE human link
24089191	Influenza attenuates IL-1β/NF-κB → impaired Type 17 defense
41927528	GSDMD pyroptosis in flu-MRSA co-infection; disulfiram rescue
42089309	Temporal type I IFN: pathogenic myeloid TNF-α, suppressed IFN-γ
11888586	Case-control defining PVL+ necrotizing pneumonia phenotype
21051855	92-case review; mortality predictors; toxin-suppression benefit
33833037	PVL (HR 1.81) & MRSA (HR 2.37) independent mortality predictors
42501530	Autopsy + review: fatal PVL necrotizing pneumonia
23524477	VAP epidemiology; incidence & attributable mortality
24709718	*S. aureus* most frequent (29%) organism in VAEs
24277023	MRSA nasal PCR NPV 99.2% — de-escalation tool
36852899	Vancomycin & linezolid guideline first-line for MRSA pneumonia
40349971	Cefazolin/ASP first-line for MSSA
33894131	Suvratoxumab anti-Hla mAb prevention (SAATELLITE)
21722847	CA-MRSA pneumonia clinical features & therapy
32919518	Complications of complicated pneumonia
23418434	LA-MRSA CC398 zoonosis
42161902	Superantigens + Hla drivers; murine models; nanobodies

Limitations and Knowledge Gaps

1. **Template mismatch.** This template is designed for genetic/rare diseases; sections on causal human genes, ACMG variant classification, inheritance, penetrance, carrier frequency, and host genetic testing are **not applicable**. The "molecular genetics" of this disease is bacterial.
2. **Human vs model evidence.** Much mechanistic detail (Hla/ADAM10, PSMα necroptosis, GSDMD pyroptosis) is from mouse/in-vitro work; human PVL biology differs (human neutrophils far more PVL-sensitive), complicating translation.
3. **PVL causal debate.** PVL is a strong clinical marker of severe necrotizing disease, but its precise causal weight vs Hla/PSMα in humans remains debated; PVL-negative necrotizing MSSA also occurs (PMID: 41585354).
4. **Epidemiologic heterogeneity.** Prevalence/mortality vary widely by region, clone, and care setting; a precise incidence per 100,000 for "*S. aureus* pneumonia" specifically is not cleanly defined in the literature reviewed.
5. **No approved vaccine or anti-toxin therapy is yet standard of care**, despite promising models and phase-2 trials.
6. **No primary experimental dataset** was analyzed here; conclusions are literature-derived.

Proposed Follow-up Experiments / Actions

1. **Human validation of anti-Hla prophylaxis:** analyze SAATELLITE and successor phase-3 outcomes stratified by baseline PVL/agr genotype and influenza co-infection status.
2. **Prospective PVL/agr-typed cohort:** quantify independent contributions of Hla vs PVL vs PSMα to human mortality, adjusting for MRSA status and influenza.
3. **Host-directed therapy trials:** test GSDMD inhibition (disulfiram) and Type 17 augmentation (IL-23/IL-17 axis) as adjuncts in severe/co-infection pneumonia.
4. **Biomarker development:** validate leukopenia, hemoptysis, and circulating toxin/ADAM10-cleaved E-cadherin fragments as early prognostic markers.
5. **Diagnostic stewardship rollout:** broaden MRSA nasal-PCR-guided de-escalation and mNGS for co-infection detection in refractory pneumonia.
6. **One-Health surveillance:** monitor LA-MRSA CC398 spillover and PVL acquisition by hospital lineages.

7. **Vaccine antigen strategy:** pursue multi-antigen/toxoid formulations (Hla + PVL + PSM + superantigens) given single-antigen failures.

Report compiled from 16 confirmed findings and 69 reviewed papers over 5 investigation iterations. Evidence types are labeled inline as human clinical (cohorts/case-control/RCT), model organism (murine/porcine), or in vitro where relevant.
